

Capture the Flag

Last mission, you built reusable Angular components for the Student Portal.

Unfortunately, you hardcoded every piece of information directly into the HTML. Every student's profile looks identical, and announcements can only be changed by editing the HTML.

Your mission is to **bring the portal to life using Angular Interpolation** (`{{ }}`).

Note from the instructor before you start:

These activities involve using your own logic. We are strictly to implement NOT TO USE AI to generate the logic for you. You may only use AI for RESEARCHING the syntax or UI, but NEVER THE LOGIC.

Consequence: Immediate 0 score in your grade for this activity.

Flag 0 – Mission Branch

Create and switch to a new branch: `feature/interpolation`

Capture

- Show your current Git branch.

```
woj@2 ITFE403A_RIZAL % ls
Angular Activity1.pdf
Copy of Activity 2_ Angular Components.pdf
woj@2 ITFE403A_RIZAL % cd my-app
woj@2 my-app % git checkout -b feature/interpolation
Switched to a new branch 'feature/interpolation'
woj@2 my-app %
```

Flag 1 – Escape the Hardcoded Trap

Open your **Student Profile Card**.

Right now, your HTML probably looks something like this:

```
<h2>Juan Dela Cruz</h2>
<p>BSIT</p>
<p>3rd Year</p>
<p>2026-0001</p>
```

Your mission is to remove the hardcoded values.

Move them into your **TypeScript** component instead. (".ts")

Hint

```
{{ variableName }}
```

Your HTML should no longer contain actual student information. It should only reference variables.

Your profile card should display:

- Student Name
- Student Number
- Course
- Year Level
- Section
- Nickname (all lowercase)
- Email
- Grade



Capture

- HTML file
- TypeScript file
- Browser

```

<div class="credentials-body">
  <div class="name-block">
    <span class="person-name">{{StudentName}}</span>
  </div>

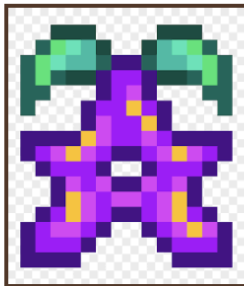
  <div class="info-rows">
    <div class="info-row">
      <span class="info-label">ID No</span>
      <span class="info-separator">:</span>
      <span class="info-value">{{StudentID}}</span>
    </div>
    <div class="info-row">
      <span class="info-label">Course</span>
      <span class="info-separator">:</span>
      <span class="info-value">{{Course}} - {{Year}}</span>
    </div>
    <div class="info-row">
      <span class="info-label">Date</span>
      <span class="info-separator">:</span>
      <span class="info-value">July 29, 2026</span>
    </div>
    <div class="info-row">
      <span class="info-label">Section</span>
      <span class="info-separator">:</span>
      <span class="info-value">{{Section}}</span>
    </div>
    <div class="info-row">
      <span class="info-label">Nickname</span>
      <span class="info-separator">:</span>
      <span class="info-value">{{Nickname}}</span>
    </div>
    <div class="info-row">
      <span class="info-label">Email</span>
      <span class="info-separator">:</span>
      <span class="info-value">{{Email}}</span>
    </div>
    <div class="info-row">
      <span class="info-label">Grade</span>
      <span class="info-separator">:</span>
      <span class="info-value">{{Grade}}</span>
    </div>
  </div>
</div>

```

```

1  import { Component } from '@angular/core';
2
3
4  @Component({
5    selector: 'app-profile-card',
6    imports: [],
7    templateUrl: './profile-card.html',
8    styleUrls: ['./profile-card.css'],
9  })
10 export class ProfileCard {
11   StudentName:string = "Kurt Wojtyle S. Rizal";
12   StudentID =23017245
13   Course:string = "BSIT";
14   Year:string = "IV";
15   Section:string = "Undefined";
16   Nickname:string = "kurt";
17   Email:string = "kurtwojtyle.rizal@dbtc-cebu.edu.ph";
18   Grade: any;
19 }

```



Authorize Signature

MY FIRST ANGULAR ACTIVITY

Kurt Wojtyle S. Rizal

ID No : 23017245
 Course : BSIT - IV
 Date : July 29, 2026
 Section : Undefined
 Nickname : kurt
 Email : kurtwojtyle.rizal@dbtc-cebu.edu.ph
 Grade :



Flag 2 – Dynamic Profile Card

Verify that **every** piece of information in your profile card now comes from TypeScript.

If changing a value in the TypeScript file updates the browser without touching the HTML, you've captured the flag.

Capture

- Updated TypeScript file with changed value
- Browser

Flag 3 – Dynamic Announcements

Your announcement component is also hardcoded.

Instead of writing directly in HTML, create variables for each announcement.

Example idea:

announcementTitle1
announcementBody1

announcementTitle2
announcementBody2

announcementTitle3
announcementBody3

Use interpolation so every announcement card gets its content from TypeScript.



Imagine these values came from a JSON file or API in the future.

Capture

- HTML file

- TypeScript file
- Browser

The screenshot shows the VS Code editor with the file explorer on the left and the editor window on the right. The file explorer shows the project structure, including the 'components' folder and the 'announcement-card' component. The editor window displays the TypeScript code for the 'announcement-card' component, which is a class that implements the 'Component' interface from '@angular/core'. The code includes a selector, imports, templateUrl, styleUrls, and a class with properties for announcement titles and bodies.

```

1 import { Component } from '@angular/core';
2
3 @Component({
4   selector: 'app-announcement-card',
5   imports: [],
6   templateUrl: './announcement-card.html',
7   styleUrls: ['./announcement-card.css'],
8 })
9 export class AnnouncementCard {
10   announcementTitle1:string = "AnnouncementTitle1";
11   announcementBody1:string = "AnnouncementBody1";
12
13   announcementTitle2:string = "AnnouncementTitle2";
14   announcementBody2:string = "AnnouncementBody2";
15
16   announcementTitle3:string = "AnnouncementTitle3";
17   announcementBody3:string = "AnnouncementBody3";
18 }
19

```

The terminal at the bottom shows the command 'woj@2 my-app % git checkout -b feature/interpolation' and the output 'Switched to a new branch 'feature/interpolation''.

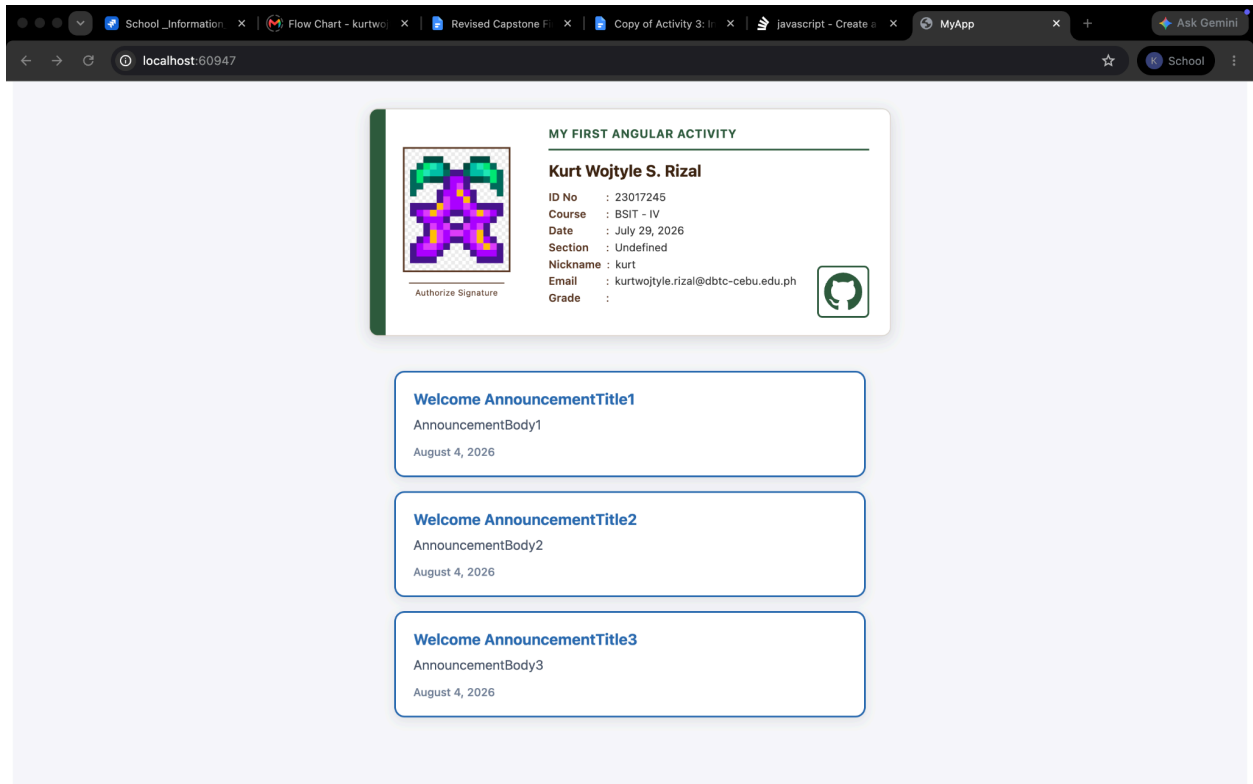
The screenshot shows the VS Code editor with the file explorer on the left and the editor window on the right. The file explorer shows the project structure, including the 'components' folder and the 'announcement-card' component. The editor window displays the HTML code for the 'announcement-card' component, which is a template for the component. The code includes a router-outlet, a main class, a profile card, a section for announcements, and a footer.

```

1 <app-header></app-header>
2
3 <main class="dashboard-body">
4   <app-profile-card></app-profile-card>
5
6   <section class="announcements-section">
7     <app-announcement-card></app-announcement-card>
8   </section>
9 </main>
10
11 <app-footer></app-footer>
12
13 <router-outlet />
14

```

The terminal at the bottom shows the command 'woj@2 my-app % git checkout -b feature/interpolation' and the output 'Switched to a new branch 'feature/interpolation''.



Flag 4 – Thinking Like Data

Now add 7 more cards to make the announcements a total of 10 cards.

Capture

- Show your ten announcement cards in Browser
- TypeScript file

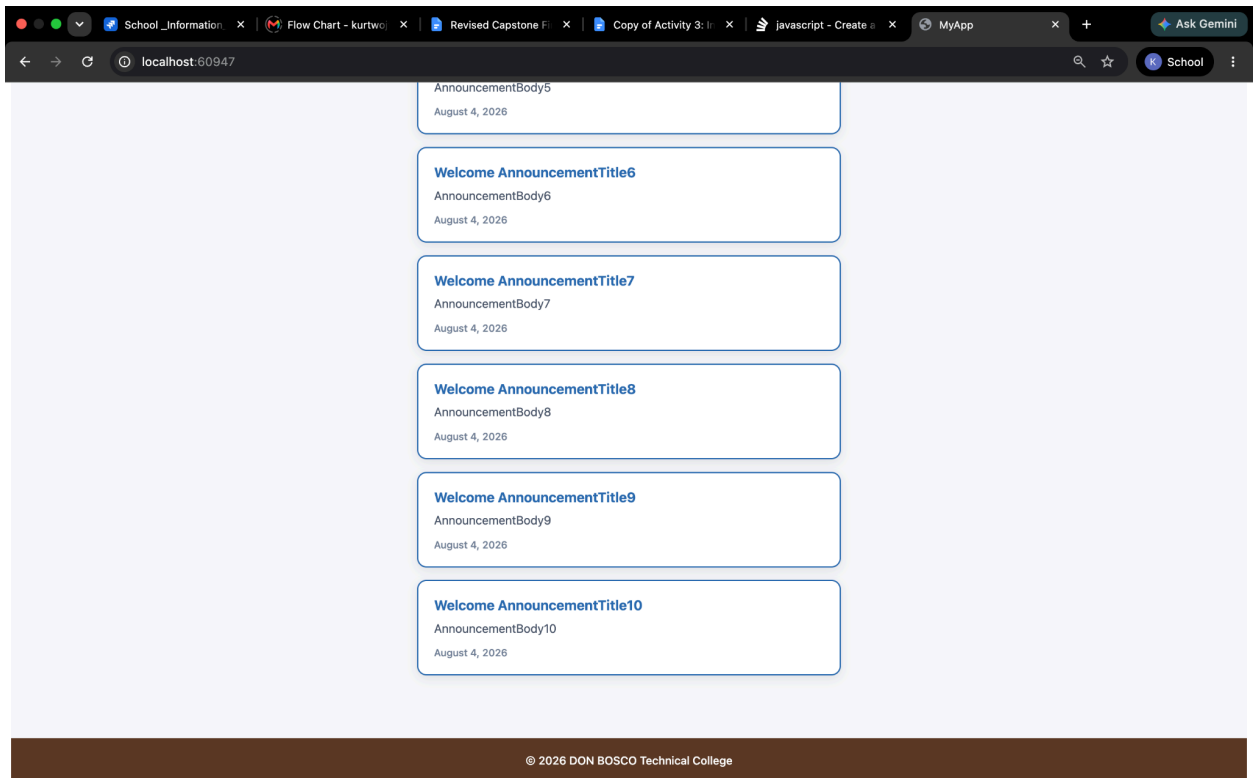
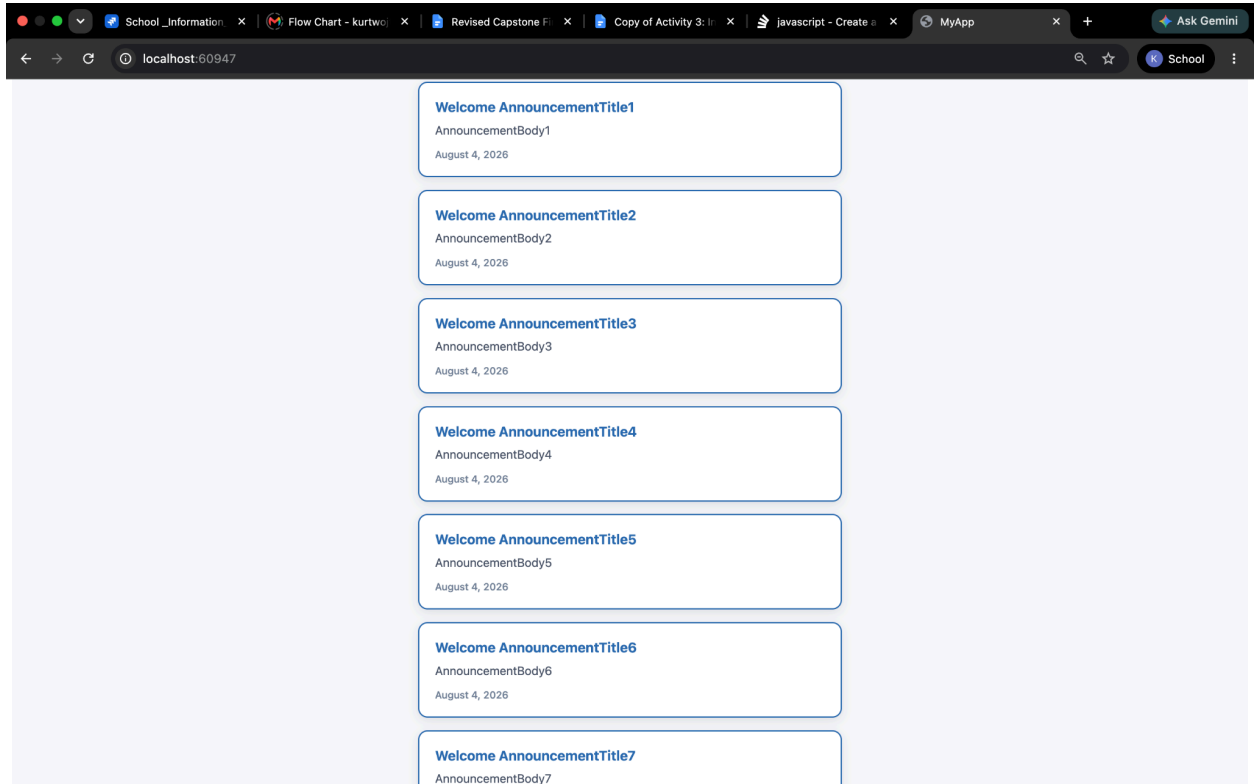
Notice how your announcement variables are starting to repeat.

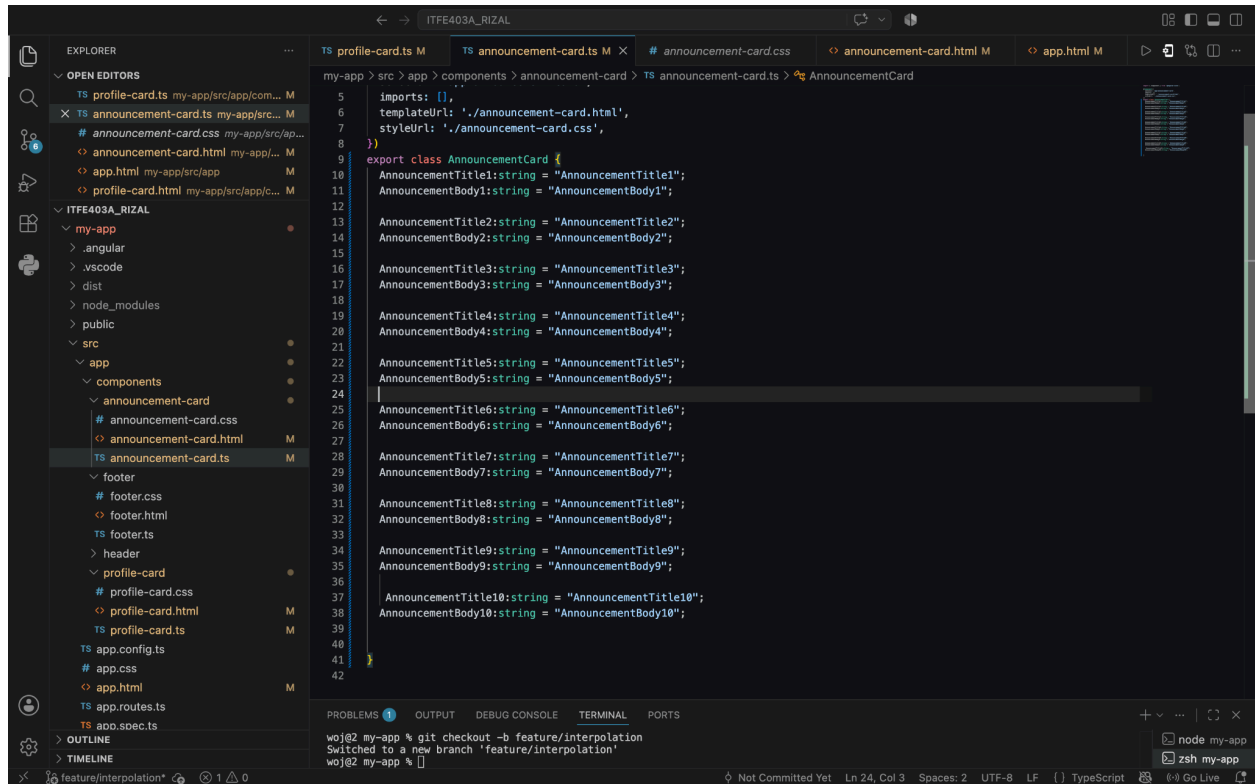
This is normal, but not ideal.

Question to think about (you don't need to answer, just think):

If there were 100 announcements, would creating 200 variables be a good idea?

(You'll solve this in a future lesson.)





Flag 5 – Academic Standing

Your profile already displays the student's grade.

Now make the portal smarter.

Below the grade, display:

- **Pass**
- **Fail**

depending on the grade.

 **Hint**

Angular interpolation can evaluate expressions.

```
{{ condition ? "True Value" : "False Value" }}
```

Grading condition: A student can only pass if their grade is 1.0 to 3.0.

Capture

- Interpolation condition
- Browser

```
</div>
| | <div class="info-row">
| |   <span class="info-label">Grade</span>
| |   <span class="info-separator">:</span>
| |   <span class="info-value">{{Grade > 3.0 ? 'Fail': 'Pass'}}</span>
| | </div>
```



Authorize Signature

MY FIRST ANGULAR ACTIVITY

Kurt Wojtyle S. Rizal

ID No : 23017245
Course : BSIT - IV
Date : July 29, 2026
Section : Undefined
Nickname : kurt
Email : kurtwojtyle.rizal@dbtc-cebu.edu.ph
Grade : Pass



Flag 6 – Formatting Text

Your student's nickname is stored in lowercase.

Example:

juan

Display it in uppercase without changing the value of the variable.

Hint

Interpolation can call simple methods.

Expected:

JUAN

Capture

- Method used to display it in uppercase
- Browser

```
import { Component } from '@angular/core';
import { UpperCasePipe } from '@angular/common';

@Component({
  selector: 'app-profile-card',
  imports: [UpperCasePipe],
  templateUrl: './profile-card.html',
  styleUrls: ['./profile-card.css'],
})
```



Authorize Signature

MY FIRST ANGULAR ACTIVITY

Kurt Wojtyle S. Rizal

ID No : 23017245
Course : BSIT - IV
Date : July 29, 2026
Section : Undefined
Nickname : KURT
Email : kurtwojtyle.rizal@dbtc-cebu.edu.ph
Grade : Pass



Flag 7 – School Email

Every student has the same format of email in Don Bosco.

Example:

juan.delacruz@dbtc-cebu.edu.ph

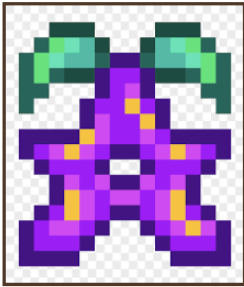
Instead of storing the full email, remove the “email” variable, then construct the email inside the template, by using the student’s name as variable, concatenated with the email suffix.

No extra variable should be created.

Capture

- Your logic to concatenate the name and email suffix using interpolation
- Browser

```
Username:string = "kurtwojtyle.rizal";
Domain:string = "@dbtc-cebu.edu.ph";
Email: string = `${this.Username}${this.Domain}`;
Grade = 1.0;
```



Authorize Signature

MY FIRST ANGULAR ACTIVITY

Kurt Wojtyle S. Rizal

ID No : 23017245
Course : BSIT - IV
Date : July 29, 2026
Section : Undefined
Nickname : KURT
Email : kurtwojtyle.rizal@dbtc-cebu.edu.ph
Grade : Pass



Flag 8 – Copyright Footer

Every website needs a footer.

Create a variable:

currentYear

Display:

© 2026 Student Portal

using interpolation.

Example idea:

© {{ currentYear }}

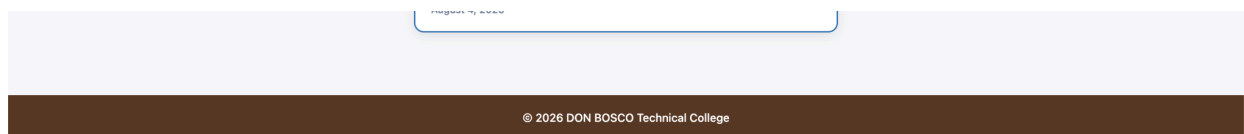


Capture

- Method used to display the currentYear without hardcoding “2026”
- Browser

```
<footer class="footer">
|  © {{currentYear}} DON BOSCO Technical College
</footer>
```

```
export class Footer {
|  currentYear: number = new Date().getFullYear();
}
```



Imagine the year changes from 2026 to 2027. If you hardcoded the year, you will have to change this value yearly. You need to have a function for that.

Flag 9 – Commit Your Progress

Final Flag Screenshot

- The entire completed browser UI for this activity
- Successful push to [feature/interpolation](#)

```
woj@2 ITFE403A_RIZAL % cd my-app
woj@2 my-app % git checkout -b feature/interpolation
Switched to a new branch 'feature/interpolation'
woj@2 my-app % git add .
woj@2 my-app % git commit -m "Activity 3: Interpolation"
[feature/interpolation 15f984c] Activity 3: Interpolation
 7 files changed, 138 insertions(+), 15 deletions(-)
woj@2 my-app % git switch -c feature/interpolation
fatal: a branch named 'feature/interpolation' already exists
woj@2 my-app % git push -u origin feature/interpolation
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 8 threads
Compressing objects: 100% (16/16), done.
Writing objects: 100% (16/16), 2.26 KiB | 2.26 MiB/s, done.
Total 16 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), completed with 6 local objects.
remote: This repository moved. Please use the new location:
remote:   https://github.com/W0jkurt/ITE403A_RIZAL.git
remote:
remote: Create a pull request for 'feature/interpolation' on GitHub by visiting:
remote:   https://github.com/W0jkurt/ITE403A_RIZAL/pull/new/feature/interpolation
remote:
To https://github.com/W0jkurt/ITFE403A_RIZAL.git
 * [new branch]      feature/interpolation -> feature/interpolation
branch 'feature/interpolation' set up to track 'origin/feature/interpolation'.
woj@2 my-app %
```